

submodules

Owned-submodule capabilities

Inventory of the discrete features each owned (own-org v a s i c - d i g i t a l / HelixDevelopment) submodule under submodules / * provides to HelixCode. Assessed from each submodule's README .md, exported package surface, and *_test .go presence (CONST-035 / §11.4.107 — evidence-based, no bluffs).

Wiring model (load-bearing). HelixCode's inner Go module (helix_code / go .mod) replaces + imports only **3** submodules directly — helix_agent (dev .helix .agent), dag_orchestrator (dev .helix .dag), pipeline_runtime (dev .helix .pipeline). HelixCode source additionally imports a handful by module path (containers, helixspecifier, helixqa, concurrency, helixmemory, debate, lazy, llmsverifier, memory, llmprovider). A large set is wired **transitively** because helix_agent / go .mod requires ~35 own-org submodules — these are reachable through the agent dependency but NOT imported by HelixCode directly (Wired=partial). The remainder are present in the repo as equal-codebase submodules (CONST-051) but are NOT in HelixCode's Go build graph at all (Wired=no). 🎥 Video=no for every row (conductor owns video confirmation; nothing here is confirmed). Origin=native throughout (all own-org).

Area	Component	Feature	Dev	Wired	Real-use	Tests
submodule	helix_agent	Ensemble multi-LLM agent service (response fusion)	done	yes	yes	unit,inte
submodule	helix_agent	ReAct/tool-calling agent runtime (consumed by HelixCode internal / agent)	done	yes	yes	unit,inte
submodule	helix_agent	Skill registry + debate + verifier wiring (umbrella, 1583 test files)	done	yes	unknown	unit,inte
submodule	dag_orchestrator	Agent-free DAG scheduler (topo dispatch, worker pool, retry/backoff)	done	yes	yes	unit
submodule	pipeline_runtime	Staged streaming dataflow runtime (push operators + FBP backpressure)	done	yes	yes	unit
submodule	helix_specifier	Spec-driven dev fusion (SpecKit 7-phase + TDD + GSD milestones)	done	partial	unknown	unit,inte
submodule	helix_specifier	10 power features (parallel exec, constitution-as-code, debate phases)	partial	partial	unknown	unit
submodule	helix_memory		done	yes	unknown	unit,inte

Area	Component	Feature	Dev	Wired	Real-use	Tests
		Unified cognitive memory (Mem0+Cognee+Letta+Graphiti fusion)				
submodule	helix_memory	Parallel cross-backend search + 3-stage fusion/dedup/re-rank	done	yes	unknown	unit
submodule	debate_orchestrator	Multi-agent debate orchestration (consensus + dissent, LessonBank)	done	yes	unknown	unit
submodule	debate_orchestrator	Aux packages (validation/audit/evaluation/reflexion/tools)	stub	partial	no	unit
submodule	concurrency	Worker pools, priority queues, rate limiters, circuit breakers, semaphores	done	yes	yes	unit
submodule	lazy	Type-safe lazy-init primitives (sync.Once generics)	done	yes	yes	unit
submodule	memory	Mem0-style scoped memory + entity extraction + knowledge graph + leak detect	done	partial	unknown	unit
submodule	llm_provider	LLMProvider interface + circuit breaker + health monitor + retry + lazy	done	partial	unknown	unit
submodule	llms_verifier	LLM verification platform (existence/latency/streaming/vision/embeddings)	done	yes	unknown	unit,inte
submodule	llms_verifier	Single-source-of-truth model/provider metadata (CONST-036/037)	done	yes	unknown	unit,inte
submodule	containers	Container lifecycle (boot/compose/health) for infra-on-demand (§11.4.76)	done	yes	unknown	unit,inte
submodule	helix_qa	Autonomous QA / Challenge orchestration (test-bank runner) [QA infra]	done	yes	unknown	unit,inte
submodule	panoptic	Recording-validator / observation harness [QA infra]	done	partial	unknown	unit,inte
submodule	agentic	Graph-based agentic workflow engine (branch/checkpoint/self-correct)	done	partial	unknown	unit
submodule	auth	JWT / API-key / OAuth2 / HTTP auth middleware / token store	done	partial	unknown	unit
submodule	cache	Multi-backend cache (mem/Redis/PG) + distributed patterns + TTL/eviction	done	partial	unknown	unit

Area	Component	Feature	Dev	Wired	Real-use	Tests
submodule	storage	Object storage (S3/MinIO/local) + cloud credential mgmt	done	partial	unknown	unit,inte
submodule	vector_db	Unified vector store (Qdrant/Pinecone/Milvus/pgvector)	done	partial	unknown	unit,inte
submodule	embeddings	Text embeddings across 7 providers (OpenAI/Cohere/Voyage/Jina/Google/Bedrock)	done	partial	unknown	unit
submodule	event_bus	Typed pub/sub bus + glob/prefix/metadata filtering + middleware	done	partial	unknown	unit
submodule	messaging	Message-broker abstraction (in-mem/Kafka/RabbitMQ) + producer/consumer patterns	done	partial	unknown	unit
submodule	streaming	SSE / WebSocket(rooms) / gRPC streaming / webhook(HMAC) / HTTP+breaker / Gin	done	partial	unknown	unit
submodule	rag	RAG pipeline (3 chunkers, BM25+semantic+hybrid retrieve, MMR rerank, RRF fusion)	done	partial	unknown	unit
submodule	tool_schema	Tool schema/validation/exec + 14 built-in safe tool handlers	done	partial	unknown	unit
submodule	plugins	Plugin lifecycle (dep-ordered) + .so/process loaders + sandbox + output parse	done	partial	unknown	unit
submodule	planning	AI planning algorithms (HiPlan + MCTS + Tree-of-Thoughts)	done	partial	unknown	unit
submodule	optimization	Semantic cache + prompt compression + structured-output + SGLang/LlamaIndex/LangChain	done	partial	unknown	unit
submodule	self_improve	RLAIF pipeline (reward models, feedback, policy/prompt optimization)	done	partial	unknown	unit
submodule	benchmark	LLM benchmarking (SWE-bench/HumanEval/MMLU/GSM8K) + leaderboard	done	partial	unknown	unit
submodule	llm_ops	LLMOps (continuous eval, A/B experiments, prompt versioning, alerting)	done	partial	unknown	unit
submodule	mcp_module	MCP server+client (JSON-RPC 2.0 stdio+HTTP/SSE) + adapter registry	done	partial	unknown	unit

Area	Component	Feature	Dev	Wired	Real-use	Tests
submodule	models	Shared AI/LLM data types (LLMRequest/Response, MCP/LSP/ACP, code-intelligence)	done	partial	unknown	unit
submodule	formatters	Pluggable code-formatter registry + engine + cache + native-binary shims	done	partial	unknown	unit
submodule	vision_engine	CV + LLM-vision UI analysis + navigation-graph construction	partial	partial	unknown	unit
submodule	doc_processor	Doc-feature-map extraction + verification-coverage tracking (QA-oriented)	done	partial	unknown	unit
submodule	security	Security tooling module (round-300 deep-doc) — needs deeper inventory	done	partial	unknown	unit,inte
submodule	red_team	YAML-driven adversarial-prompt fixture harness (defensive guardrail regression)	done	partial	unknown	unit
submodule	llm_orchestrator	Headless CLI-agent orchestration (OpenCode/Claude/Gemini/Junie/Qwen)	done	no	no	unit
submodule	helix_llm	Distributed LLM service (OpenAI/Anthropic APIs, llama.cpp, RAG, ReAct, HTTP/3)	done	no	no	unit,inte
submodule	conversation	Infinite-context (event-sourcing) + LLM compression + LRU cache	done	no	no	unit
submodule	database	Driver-agnostic DB (PG/SQLite) + pool + migrations + repo + query builder	done	no	no	unit,inte
submodule	document	Document model (18-format detect, change tracking, JSON serialize)	done	no	no	unit
submodule	filesystem	Multi-protocol FS client (SMB/FTP/NFS/WebDAV/Local)	done	no	no	unit
submodule	config	Config mgmt (JSON files, env binding, validation, 8 storage-protocol types)	done	no	no	unit
submodule	i18n	i18n library (bundles, loader, HTTP middleware)	done	no	no	unit
submodule	observability	Distributed tracing + metrics + structured logging + health + analytics	done	no	no	unit
submodule	middleware		done	no	no	unit

Area	Component	Feature	Dev	Wired	Real-use	Tests
		Reusable net/http middleware (requestid/logging/recovery/cors/chain)				
submodule	rate_limiter	Rate limiting (in-mem + Redis) + HTTP middleware	done	no	no	unit,inte
submodule	recovery	Named circuit breakers + periodic health checks + resilience facade	done	no	no	unit
submodule	watcher	Filesystem change monitoring (debounce, filters, handler chains)	done	no	no	unit
submodule	background_tasks	Persistent task queue (PG) + worker pool + stuck-detect + DLQ + progress	done	no	no	unit
submodule	skill_registry	Skill mgmt (load YAML/JSON/MD, register, execute, validate, store)	done	no	no	unit
submodule	embeddings	(see embeddings row — module also standalone)	—	—	—	—
submodule	auto_temp	LLM temperature auto-tuning (multi-temp run + multi-judge scoring)	done	no	no	unit
submodule	hyper_tune	LLM hyperparameter optimization (random/grid/Bayesian-lite)	done	no	no	unit
submodule	i_llm	Structured-reasoning patterns (CoT/ToT/ReAct/few-shot/prompt-chain)	done	no	no	unit
submodule	veritas	AI-truthfulness verification, fact-check, hallucination detection	done	no	no	unit
submodule	leak_hub	System-prompt-leak detection + searchable archive	done	no	no	unit
submodule	claritas	Defensive guardrail (leaked-prompt archive + extraction-attempt detector)	done	no	no	unit
submodule	gandalf_solutions	Read-only Gandalf prompt-hacking solutions archive (research)	done	no	no	unit
submodule	ouroborous	Self-referential AI-safety (recursive self-improve, runaway-loop detect)	done	no	no	unit
submodule	normalize		done	no	no	unit

Area	Component	Feature	Dev	Wired	Real-use	Tests
		Adversarial-input canonicalization (base64/leet/homoglyph/NFKC/ROT13...)				
submodule	plinius_common	Plinius shared lib (config validators, error types, gRPC client, i18n, types)	done	no	no	unit
submodule	toon	Token-Oriented Object Notation encode/decode	stub	no	no	unit

55 features across 50 submodules.

Coverage-depth honesty

- **Shallow-inventoried (principal features only; deeper inventory warranted):** `helix_agent` (1583 test files — only top umbrella capabilities captured; internal package-level features not enumerated), `helix_qa` (646 test files, QA infra — single capability row), `helix_specifier` (37 tests, 10 power features collapsed to 2 rows), `security` (deep-doc module — only a single umbrella row; per-package security capabilities NOT enumerated, flagged as needs-deeper-inventory), `panoptic` (QA infra — single row).
- **Real-use** is unknown for nearly every transitively-wired (`partial`) and standalone (`no`) submodule because reachability through `helix_agent`'s `go.mod` $\neq$ proof an end user exercises the capability via `HelixCode`; only the 3 directly-replaced modules + concurrency/lazy show yes.
- **Dev reflects README claims + test presence**, NOT runtime verification. `toon` is honestly `stub` (`PENDING_IMPLEMENTATION` per its own README); `debate_orchestrator` aux packages are `stub` (`NotYetImplemented` per `ACK-STUB`).
- **Tests column** marks `unit` baseline (all 70 own-org submodules have `*_test.go`); `integ/e2e` added only where README/dir evidence shows real infra/integration suites (`cache`, `storage`, `vector_db`, `llms_verifier`, `helix_qa`, `helix_agent`, `containers`, `database`, `rate_limiter`, `helix_llm`, `observability-N/A`).
- Infra/tooling submodules excluded from feature rows per task scope: `docs_chain`, `challenges` (and `containers/helix_qa/panoptic` included only as the capabilities `HelixCode` consumes, marked QA/infra).